A gym recorded 10 members. Each row is one member. The goal is to predict whether a member Renewed. (Made-up data)

Member ID	Age	Plan	Favorite Class	Visits Per Month	Training Hours Per Month	Monthly Fee ($)	Years as Member	Uses Locker	Renewed
M01	22	Basic	Yoga	8	0	20	1	No	Yes
M02	34	Plus	Spin	14	2	35	3	Yes	Yes
M03	51	Premium	Boxing	10	4	50	5	Yes	Yes
M04	28	Basic	Spin	6	0	20	1	No	No
M05	45	Plus	Yoga	12	1	35	4	Yes	Yes
M06	19	Basic	Boxing	16	0	20	1	No	Yes
M07	39	Premium	Yoga	18	40	50	6	Yes	Yes
M08	62	Plus	Yoga	9	3	35	8	No	Yes
M09	26	Plus	Spin	20	2	35	2	Yes	Yes
M10	33	Basic	Boxing	4	0	20	2	No	No

Quiz 5

Dataset 7 (gym): prepare these columns for a model.

1. **Plan is an ordered label and Favorite Class is an unordered label. Name the right encoding for each. Then write the encoded Plan for M01, M03 and M05 (use Basic = 0, Plus = 1, Premium = 2) and the one-hot columns (Yoga, Spin, Boxing) for M02.**
2. **Bin Age into Under 30, 30 to 49, and 50 or older. Give the bin for M03, M05 and M08, and say how many members are in each bin.**
3. **Min-max scale Visits Per Month (smallest 4, largest 20). Write the formula and calculate the scaled value for M01, M05 and M09.**
4. **Min-max scale Training Hours Per Month for M02, M03 and M07. What happens to most members, and what would you do about it?**
5. **Monthly Fee and Plan carry the same information. Show this using the data, and say which of the two you would drop. Name one more column you would drop.**